

FAITH OGUNWOYE

[LinkedIn](#) | [CAD design](#) | [Github](#) | ogunwoyefaito@gmail.com

SUMMARY

Mechanical Engineer and IoT research student with experience in intelligent sensing, machine learning for physical systems, and simulation-driven design. Background in CAD/CAM, FEA, and computational modeling.

EDUCATION

Skolkovo Institute of Science and Technology (Skoltech) *September 2025 – Present*
MSc. Next Generation Communication Technologies

- Specialization in IoT, Robotics and CAD/Prototyping

Abiola Ajimobi Technical University, Nigeria *January 2018 – February 2024*
BSc. Mechanical Engineering (GPA: 4.77 / 5)

- Thesis Title: Design, Fabrication and Performance Evaluation of Smart Fuel Screening and Engine Protection Device.

TECHNICAL KNOWLEDGE

Languages – Python, HTML/CSS, Solidity

Stack – Scikit-Learn, Django REST framework

Tools – ROS2, RViZ, Gazebo, Git, Fusion 360, Unity, MATLAB

Laboratory – 3D Printing, CNC Lathe, Mechanical Materials Test, Programmable Logic Controllers

RESEARCH EXPERIENCE

VLA Mobile Robot Manipulation, Skoltech *2026*
Supervisor: Prof. Dzmitry Tsetserukou

- Operator for real-world dataset collection using dual-arm teleoperation for complex pick-and-place tasks (**egg from crate, can/waffles to box**) partnering with simulation, SLAM, and mechanics teams to generate high-quality demonstration data, enabling robot learning for successful inference

Digital Twin, Automation and Control Group *2021 – 2022*
Research Assistant

- Designed a smart automatic solar tracking system using LDR-based sensing and control logic, 42% energy efficiency increase when compared to a fixed axis solar panel.

Abiola Ajimobi Technical University *2020 – 2021*
£155,464 U.K. Royal Academy of Engineering Project: “AI for Clean Energy (AI4CE)”

- Analyzed clean energy generation datasets using AI-based methods to study performance trends and variability across operating conditions.

PROJECTS

ML Model Robustness Analysis – Coded Compressive Sensing *October 2025 – December 2025*
• Built a compressive sensing framework for decoding simultaneous wireless transmissions from 100+ users; achieved error rate $P_e \leq 0.05$, outperforming a $4\times$ ALOHA baseline.

Adaptive Antenna Beamforming System for MU-MIMO *October 2025 – December 2025*
• Developed a learnable beamforming architecture for 64-antenna arrays (ULA vs UCA) serving 8 simultaneous users; validated across 100 channel realizations with measurable SNR gains over fixed baselines.

LDR(Light Dependent Resistor)-Based Automatic Solar Tracking System *November 2022*

- Contributed to design analysis and digital prototyping; produced full 3D CAD models and animations in Fusion 360 to support stakeholder communication and system visualization.

Bioreactor Design and CFD Analysis

February 2025

- Designed a lab-scale bioreactor optimized for flow uniformity; used CFD simulation to identify and eliminate dead zones, informing design decisions prior to fabrication.

Genotype / Sample Processing Machine

April 2023

- Produced a fully parameterized 3D CAD assembly incorporating modular components.

PUBLICATIONS

- **PointDiffusion: Diffusion-Based Scene Completion in the Point Cloud Domain.** 2026 [Link](#)
- **HapticVLA: Contact-Rich Manipulation via Vision-Language-Action Model without Inference-Time Tactile Sensing.** 2026 [Link](#)
- **Prediction of Impact Strength of TIG Welded Cr-Mo Steel Using Artificial Neural Network.** 2024 [Link](#)
- **Exploring the impact of diesel-vegetable oil blends as an alternative fuel in combustion chambers.** 2024 [Link](#)
- **Creating New Materials and Technologies to Reduce the Environmental Impact of Human Activity.** 2023 [Link](#)
- **Predictive maintenance of industrial equipment using machine learning in industrial environment of Awka Metropolis, Nigeria.** 2023 [Link](#)
- **Design and Fabrication of a Mechanical Sweeper.** 2024 [Link](#)

PROFESSIONAL EXPERIENCE

Mechanical Engineer

March 2025 – August 2025

Dangote Fertilizer Ltd, Nigeria

- Interned at Africa's leading fertilizer producer performing maintenance and optimization of machinery.

Graduate Assistant

April 2023 – March 2024

University of Ilorin, Nigeria

- Planned lectures, seminars, and graded students' coursework and project work.

University CAD Instructor

August 2022 – October 2022

- Handled laboratory on Machine Design, 3D Printing and CNC Machining as an instructor.

Automation Engineer

August 2021 – November 2021

- Programmed PLCs and Developed Python Django REST Framework programs

LEADERSHIP AND VOLUNTEER EXPERIENCE

- **Project Executive**, Nigerian Innovative Engineers, Scientist and Applied Technologists (2023 – 2024)
- **Leadership Volunteer**, Aspire Leaders Program, Harvard-University Initiative (2022)
- **Leadership Volunteer**, Millennium Campus Fellowship, United Nations Academic Impact (2022)
- **Course Tutor**, Nigerian Institute of Mechanical Engineers Student (2019 – 2021)

HONOURS AND AWARDS

- Graduated Top 1% in the Department of Mechanical Engineering (2023)
- Overall Best Graduating Student in French Diploma (2023)